

Supplementary material for “Searching for the digital dividend: Internet diffusion and productivity growth in 149 economies, 1996–2025”

Quang-Vinh Dang

^a*British University Vietnam, Hung Yen, Vietnam*

Appendix A. Proofs

Throughout, the year effect is a function of the time index t , which is an element of F_{it} , so λ_t is absorbed by ℓ . The two-way operator $\mathcal{M}$ is the orthogonal projection onto the complement of the space spanned by country and year indicators; it is symmetric and idempotent, and $\mathcal{M}c = 0$ for any vector with elements $c_{it} = c_i + c_t$. For country i , collect its T_i observations into Y_i , $\Phi_i (T_i \times K)$ and F_i , and write $V_i = \Phi_i - m_0(F_i)$ and $e_i = Y_i - \ell_0(F_i) - V_i\beta_0$. Under Assumptions 1–2 and $r_K = 0$, $e_{it} = a_i + \varepsilon_{it}$. Let $Q_i = I_{T_i} - T_i^{-1}\iota\iota'$ be the country-demeaning matrix, $Q = \text{diag}(Q_1, \dots, Q_G)$, and let A be the $n \times \mathcal{T}$ matrix of indicators of the $\mathcal{T}$ calendar years, with country blocks A_i . By the Frisch–Waugh–Lovell theorem the two-way operator is, for balanced and unbalanced panels alike,

$$\mathcal{M} = Q - QA(A'QA)^+A'Q,$$

where $^+$ denotes the Moore–Penrose inverse ($A'QA$ has rank $\mathcal{T} - 1$ because $A\iota$ lies in the span of the country indicators). The alternating-projection algorithm used in the code converges to this orthogonal projection (the von Neumann–Halperin theorem; see Gaure, 2013). The following lemma allows us to replace $\mathcal{M}$ by Q in the asymptotic arguments.

Assumption A.1 (Panel structure). (a) *The number of calendar years $\mathcal{T}$ is fixed.* (b) *The set of years in which country i is observed, $\mathcal{S}_i$, is independent across countries, and*

$\mathbb{E}[\varepsilon_{it} \mid \mathcal{S}_i, D_i, X_i, \alpha_i] = 0$ and $\mathbb{E}[D_{it} \mid \mathcal{S}_i, F_{i1}, \dots, F_{iT_i}] = \mathbb{E}[D_{it} \mid F_{it}]$, i.e. the unbalancedness is ignorable given the observed paths. (c) The smallest non-zero eigenvalue of $G^{-1}A'QA$ is bounded away from zero with probability approaching one; this holds, for example, if each calendar year is observed, together with at least one other year, for a fixed positive fraction of the countries.

Lemma A.1 (Year demeaning). *Let W and E be n -vectors (or $n \times K$ arrays) with country blocks W_i, E_i that are independent across countries, have bounded fourth moments, and satisfy $\mathbb{E}[A'_i Q_i W_i] = 0$ and $\mathbb{E}[A'_i Q_i E_i] = 0$. Under Assumption A.1, $G^{-1/2}W'ME = G^{-1/2} \sum_i W'_i Q_i E_i + O_p(G^{-1/2})$ and $G^{-1}W'ME = G^{-1} \sum_i W'_i Q_i E_i + O_p(G^{-1})$.*

Proof. By the expression for $\mathcal{M}$,

$$G^{-1/2}W'ME = G^{-1/2}W'QE - (G^{-1/2}W'QA)(G^{-1}A'QA)^+(G^{-1/2}A'QE)G^{-1/2}.$$

The vector $G^{-1/2}A'QE = G^{-1/2} \sum_i A'_i Q_i E_i$ is a normalised sum of G independent mean-zero terms with bounded variance, hence $O_p(1)$ by Chebyshev's inequality, and the same holds for $G^{-1/2}A'QW$. By Assumption A.1(c), $\|(G^{-1}A'QA)^+\| = O_p(1)$; because the Moore–Penrose inverse is applied on the row space of $A'Q$, in which both $A'QW$ and $A'QE$ lie, no further condition is needed for the rank-deficient direction. The second term is therefore $O_p(1) \cdot O_p(1) \cdot O_p(1) \cdot G^{-1/2} = O_p(G^{-1/2})$. The second statement follows in the same way. $\square$

In the applications of the lemma, W is $V = \Phi - m_0(F)$ or a nuisance error, and E is e or a nuisance error. The condition $\mathbb{E}[A'_i Q_i V_i] = 0$ holds because $Q_i A_i$ is a function of the observed years, which are coordinates of F , and $\mathbb{E}[V_{it} \mid \mathcal{S}_i, F_{i1}, \dots, F_{iT_i}] = 0$ by Assumptions 3 and A.1(b); $\mathbb{E}[A'_i Q_i e_i] = \mathbb{E}[A'_i Q_i \varepsilon_i] = 0$ by Assumptions 1 and A.1(b). For the nuisance errors, which are estimated on other countries and are therefore fixed functions conditionally on the training folds, the same algebra shows that the correction term is bounded by $O_p(1)$ times the product of their L_2 norms and $G^{-1/2}\|A'QV\|$ or $G^{-1/2}\|A'Qe\|$, which is of smaller order than the terms already controlled in the proof of Proposition 3.

Proof of Proposition 1. (i) For directions $\delta_\ell(F)$ and $\delta_m(F)$, and using Lemma A.1,

$$\psi(\beta_0; \ell_0 + r\delta_\ell, m_0 + r\delta_m) = \mathbb{E}[(V_i - r\delta_{m,i})' Q_i (e_i - r\delta_{\ell,i} + r\delta_{m,i}\beta_0)].$$

Differentiating at $r = 0$ gives $-\mathbb{E}[\delta'_{m,i} Q_i e_i] + \mathbb{E}[V_i' Q_i (\delta_{m,i}\beta_0 - \delta_{\ell,i})]$. In the first term, $Q_i e_i = Q_i \varepsilon_i$ because a_i is constant within country, and $\mathbb{E}[\varepsilon_{it} \mid F_{i1}, \dots, F_{iT_i}] = 0$ by Assumption 1, because F_{is} is a function of (D_i, X_i) ; the term is zero. In the second term, $Q_i w_i$ is a function of $(F_{i1}, \dots, F_{iT_i})$ for $w_i = \delta_{m,i}\beta_0 - \delta_{\ell,i}$, and, when Z_{it} is in F_{it} , $\mathbb{E}[V_{it} \mid F_{i1}, \dots, F_{iT_i}] = b(Z_{it})\{\mathbb{E}[D_{it} \mid F_{i1}, \dots, F_{iT_i}] - \mu(F_{it})\} = 0$ by Assumption 3 (in the dose-response model the same condition is imposed on Φ); the term is zero.

(ii) $\mathcal{M}(Y - \ell - c) = \mathcal{M}(Y - \ell) - \mathcal{M}c = \mathcal{M}(Y - \ell)$ whenever $c_{it} = c_i + c_t$, because $\mathcal{M}$ annihilates the span of the country and year indicators; the same holds for each column of m . If instead $\hat{\mu} = \mu + c_i$, the regressors become $\mathcal{M}\{b(Z)(D - \mu)\} - \mathcal{M}\{b(Z_{it})c_i\}$. If $b(Z_{it}) = \bar{b}_i$ does not vary within country (in particular if $K = 1$, where $b \equiv 1$, or if Z is time-invariant), then $b(Z_{it})c_i = \bar{b}_i c_i$ is a country-specific constant vector and $\mathcal{M}\{b(Z_{it})c_i\} = 0$. When $b(Z_{it})$ varies within country the term is in general non-zero. Note that multiplication by c_i does not commute with the two-way operator, $\mathcal{M}\{c_i b(Z_{it})\} \neq c_i \mathcal{M}\{b(Z_{it})\}$ in general, because $\mathcal{M}$ also removes year means; for example, with two countries observed in three years, $c = (1, 1, 1, 3, 3, 3)'$ and $b = (0, 1, 4, 1, 3, 8)'$, $\mathcal{M}(cb) = (3.67, 1.17, -4.83, -3.67, -1.17, 4.83)'$ whereas $c \odot \mathcal{M}b = (0.67, 0.17, -0.83, -2.00, -0.50, 2.50)'$. $\square$

Proof of Proposition 2. If Z_{it} is a coordinate of F_{it} , then $b_k(Z_{it})$ is F_{it} -measurable and $\mathbb{E}[D_{it} b_k(Z_{it}) \mid F_{it}] = b_k(Z_{it}) \mathbb{E}[D_{it} \mid F_{it}]$. Consequently $\Phi_{it} - m(F_{it}) = b(Z_{it})\{D_{it} - \mu(F_{it})\}$. $\square$

Proof of Proposition 3. Write $\tilde{\Phi} = \mathcal{M}(\Phi - \hat{m})$, $\tilde{Y} = \mathcal{M}(Y - \hat{\ell})$, and let $\tilde{\Phi}^0$ be the same object at the true nuisance functions. From (3),

$$\sqrt{G}(\hat{\beta}_\lambda - \beta_K) = \left(\frac{1}{G} \tilde{\Phi}' \tilde{\Phi} + \lambda \frac{\kappa}{G} \Omega \right)^{-1} \left\{ \frac{1}{\sqrt{G}} \tilde{\Phi}' (\tilde{Y} - \tilde{\Phi} \beta_K) - \sqrt{G} \lambda \frac{\kappa}{G} \Omega \beta_K \right\}.$$

The last term vanishes because $\kappa/G = O_p(1)$ and $\lambda_G \sqrt{G} \rightarrow 0$. Let $\Delta_m = \hat{m} - m_0$ and $\Delta_\ell = \hat{\ell} - \ell_0$ on the evaluation fold. By Lemma A.1,

$$\frac{1}{\sqrt{G}} \tilde{\Phi}'(\tilde{Y} - \tilde{\Phi}\beta_K) = \frac{1}{\sqrt{G}} \sum_i V_i' Q_i e_i + R_1 + R_2 + R_3 + O_p(G^{-1/2}),$$

where $R_1 = -G^{-1/2} \sum_i \Delta_{m,i}' Q_i e_i$, $R_2 = G^{-1/2} \sum_i V_i' Q_i (\Delta_{m,i} \beta_K - \Delta_{\ell,i})$ and $R_3 = -G^{-1/2} \sum_i \Delta_{m,i}' Q_i (\Delta_{m,i} \beta_K - \Delta_{\ell,i})$. Because the nuisance functions are estimated on countries outside the evaluation fold, conditionally on the training sample the summands of R_1 and R_2 are independent across countries with mean zero (by Assumptions 1 and 3, as in the proof of Proposition 1), and their conditional variances are bounded by a constant times $\|\Delta_m\|_2^2$ and $\|\Delta_m\|_2^2 + \|Q\Delta_\ell\|_2^2$; hence $R_1, R_2 = o_p(1)$. By the Cauchy–Schwarz inequality, $\|R_3\| \leq C\sqrt{G} \|\Delta_m\|_2 (\|\Delta_m\|_2 + \|Q\Delta_\ell\|_2)$, which is $o_p(1)$ under Assumption 4(b); by Proposition 2, $\|\Delta_m\|_2 \leq C\|\hat{\mu} - \mu\|_2$. Only the country-demeaned error $Q\Delta_\ell$ appears because Q annihilates country constants. Similarly $G^{-1} \tilde{\Phi}' \tilde{\Phi} \xrightarrow{p} J$. The Lindeberg–Feller central limit theorem for independent clusters gives $G^{-1/2} \sum_i V_i' Q_i e_i = G^{-1/2} \sum_i s_i \xrightarrow{d} N(0, \Omega_s)$. Consistency of $G\hat{V}$ follows from standard arguments for cluster-robust variance estimators (Cameron and Miller, 2015). For a fixed number S of splits, each $w'\hat{\beta}_s$ has the same linear representation, so the median does too and the variance adjustment is asymptotically negligible (Chernozhukov et al., 2018). $\square$

Proof of Proposition 4. Let $A_\lambda = \tilde{\Phi}' \tilde{\Phi} + \lambda \kappa \Omega$. From (3), $\hat{\beta}_\lambda = A_\lambda^{-1} \tilde{\Phi}' \tilde{Y}$, so that $\hat{\beta}_\lambda - \tilde{\beta}_\lambda = A_\lambda^{-1} \tilde{\Phi}' (\tilde{Y} - \tilde{\Phi} \beta_K)$. The expansion of $G^{-1/2} \tilde{\Phi}' (\tilde{Y} - \tilde{\Phi} \beta_K)$ in the proof of Proposition 3 does not involve the penalty, and $G^{-1} A_\lambda \xrightarrow{p} J_\lambda$, which is positive definite because J is. The first claim follows. If $\theta(z) = a + bz$, then θ lies in the span of the cubic B-spline basis, $r_K = 0$, and $b(z)' \beta_K$ has zero second derivative, so $\beta_K' \Omega \beta_K = 0$; because Ω is positive semi-definite, $\Omega \beta_K = 0$ and $\tilde{\beta}_\lambda = A_\lambda^{-1} (A_\lambda - \lambda \kappa \Omega) \beta_K = \beta_K$. The residuals $\hat{u}$ then differ from the true within-transformed errors only by terms controlled in the proof of Proposition 3, so $G\hat{V}$ is consistent. The linearity statistic is a continuous function of $\sqrt{G} ML_g(\hat{\beta}_\lambda - \beta_K)$, where

L_g evaluates $b(\cdot)$ on the grid and M removes the linear fit on the grid, and $ML_g\beta_K = 0$ under the null; its limit distribution is that of the maximum of a Gaussian vector with the estimated covariance, which is what the simulated critical value approximates. A constant θ is a special case of a linear one. $\square$

Appendix B. Data and implementation details

Variables. Main panel (1996–2025). From the World Development Indicators (July 2026 release, retrieved through the World Bank API in September 2026): GDP per person employed at constant 2021 PPP dollars (SL.GDP.PCAP.EM.KD); individuals using the internet (IT.NET.USER.ZS); mobile cellular (IT.CEL.SETS.P2) and fixed broadband (IT.NET.BBND.P2) subscriptions per 100 people; gross capital formation (NE.GDI.TOTL.ZS), government final consumption (NE.CON.GOVT.ZS), exports (NE.EXP.GNFS.ZS) and imports (NE.IMP.GNFS.ZS) as percentages of GDP; population growth (SP.POP.GROW) and total population (SP.POP.TOTL); the urban population share (SP.URB.TOTL.IN.ZS); the age-dependency ratio (SP.POP.DPND); GDP per capita at constant 2015 dollars (NY.GDP.PCAP.KD); the Statistical Performance Indicators overall score (IQ.SPI.OVRL, country mean over 2016–2024); fuel and ores-and-metals exports as percentages of merchandise exports (TX.VAL.FUEL.ZS.UN, TX.VAL.MMTL.ZS.UN; resource exporters are countries whose combined share averages at least 50% over 1996–2025); fixed telephone subscriptions per 100 people in 1990 (IT.MLT.MAIN.P2); and learning-adjusted years of schooling (HD.HCI.LAYS, country mean). Mean years of schooling is taken from the UNDP Human Development Report 2025 composite-indices time series (1990–2023); the 2024 value is the 2023 value plus the country’s 2019–2023 linear trend. Initial log GDP per worker is the average of the available values in 1991–1995 (for Montenegro, the first available value, 1997). Percentages are divided by 100. Broadband subscriptions are set to zero in the years before a country’s first reported value, on the assumption that the service did not exist; countries that never report broadband are left missing, and a robustness check leaves all

pre-report years missing. Regional and income aggregates are excluded. Region-by-year effects use the seven World Bank regions.

Screening of adoption series. Values are set to missing when the internet share is outside $[0, 100]$ or changes by more than 25 points in a year, when mobile subscriptions exceed 250 per 100 people or change by more than 100 per 100, or when broadband subscriptions exceed 60 per 100 or change by more than 20 per 100. The rules flag 39 values (8 internet, 18 mobile, 13 broadband). Only the internet flags affect the baseline sample. The list is part of the replication package. The unscreened sample, built with the same trimming cutoffs and countries as the baseline (4,081 observations), gives a headline estimate of -2.75 (1.27). The untrimmed sample keeps the baseline countries and all their observations (4,157).

PWT panels. From the Penn World Table 11.0 (1996–2023) and 10.0 (1996–2019), with identical definitions: labour productivity growth is the log change of real GDP at constant national prices (**rgdpna**) divided by persons engaged (**emp**); initial productivity is the 1991–1995 average of log output-side real GDP at current PPPs (**cgdpo**) per person engaged, and the national-prices alternative uses **rgdpna**; TFP growth is the log change of **rtfpna**; the human capital index is **hc**; the government share is **cs_h_g**; openness is **cs_h_x**–**cs_h_m**, which equals exports plus imports as shares of GDP because PWT records **cs_h_m** with a negative sign; the labour share is **labsh**. The adoption series are the ITU data distributed by Our World in Data. All controls and the treatment enter with a one-year lag in all panels.

Invariance check. We add country-specific constants to the cross-fitted residuals of the baseline and re-solve the final stage with the penalty held at its selected value (`code/invariance_check.py`). Constants added to the outcome residuals (standard deviation 1) leave every estimate unchanged up to numerical precision, as Proposition 1(ii) states. Constants added to the treatment residuals (standard deviation 0.05, against a residual standard deviation of 0.067) leave the constant-effect estimate unchanged. They also leave the Panel B estimates unchanged, because initial productivity is time-invariant, so that $b(Z_{it})$ is constant within country. For the time-varying schooling moderator they

move the tercile contrast from -0.94 to -0.33 , about 0.18 of its standard error.

Sample. Table B.1 shows how the baseline sample is obtained from the WDI country-years. Montenegro, which has no productivity data before 1997, is the only country whose initial productivity is not a 1991–1995 average; it takes its first available value, and excluding it leaves the results unchanged (Table C.6).

Table B.1: Sample construction, WDI panel.

Step	Observations	Countries
Country-years 1996-2025 (WDI economies)	6,510	217
Outcome observed	5,257	177
Treatment observed	5,042	177
Schooling observed	4,830	174
Government share observed	4,344	166
Openness observed	4,314	165
Population growth observed	4,314	165
Dependency ratio observed	4,314	165
Urban share observed	4,314	165
Initial productivity observed	4,314	165
Outcome within 1st-99th percentiles	4,226	165
Countries with at least 15 years (baseline)	4,075	149
Common sample of the earlier version	3,787	141

Notes: each row applies one additional condition to the previous row. “Observed” refers to the variable entering the regression (lagged where applicable). The last row is the sample of the earlier version of this paper, which also required mobile and broadband subscriptions, the investment share, lagged productivity and growth, and the second lag of the treatment, with trimming cutoffs computed on that sample.

Countries. Table B.2 lists the 149 countries of the main panel by tercile of initial GDP per worker.

Implementation. The nuisance learners are LightGBM gradient-boosted trees (Ke et al., 2017) with 15 leaves, a minimum of 20 observations per leaf (10 in the simulations), row and column subsampling of 80% and a learning rate of 0.03; the random-forest check uses 400 trees (Breiman, 2001) with a minimum leaf size of 5 and half of the features at each split. The feature set contains the lagged controls, their country means (except for time-invariant variables), the country means of the treatment and the moderator, and the calendar

Table B.2: Countries in the main estimation sample, by tercile of initial GDP per worker.

Group	Countries
Low	Albania; Armenia; Azerbaijan; Bangladesh; Benin; Bolivia; Bosnia and Herzegovina; Burkina Faso; Cabo Verde; Cambodia; Cameroon; Central African Republic; Chad; China; Comoros; Congo, Dem. Rep.; Equatorial Guinea; Gambia, The; Georgia; Ghana; Guinea; Guinea-Bissau; Haiti; Honduras; India; Indonesia; Kenya; Kyrgyz Republic; Lao PDR; Lesotho; Madagascar; Mali; Mongolia; Mozambique; Nepal; Nicaragua; Niger; Pakistan; Philippines; Rwanda; Senegal; Sierra Leone; Solomon Islands; Sri Lanka; Tajikistan; Tanzania; Timor-Leste; Togo; Tonga; Uganda; Uzbekistan; Viet Nam; Zambia
Middle	Angola; Belarus; Belize; Botswana; Brazil; Bulgaria; Chile; Colombia; Congo, Rep.; Costa Rica; Croatia; Dominican Republic; Ecuador; Egypt, Arab Rep.; El Salvador; Estonia; Fiji; Guatemala; Iraq; Kazakhstan; Korea, Rep.; Latvia; Lebanon; Lithuania; Malaysia; Mauritania; Mauritius; Moldova; Morocco; Namibia; North Macedonia; Panama; Paraguay; Peru; Poland; Romania; Sao Tome and Principe; Serbia; Slovak Republic; South Africa; Syrian Arab Republic; Thailand; Tunisia; Turkiye; Turkmenistan; Ukraine; Uruguay; West Bank and Gaza; Zimbabwe
High	Algeria; Argentina; Australia; Austria; Bahamas, The; Bahrain; Belgium; Brunei Darussalam; Canada; Cyprus; Czechia; Denmark; Finland; France; Gabon; Germany; Greece; Hong Kong SAR, China; Hungary; Iceland; Iran, Islamic Rep.; Ireland; Israel; Italy; Japan; Kuwait; Libya; Luxembourg; Malta; Mexico; Montenegro; Netherlands; New Zealand; Norway; Oman; Portugal; Qatar; Russian Federation; Saudi Arabia; Singapore; Slovenia; Spain; Sweden; Switzerland; United Arab Emirates; United Kingdom; United States

year. Knots are placed at equally spaced quantiles of the moderator between its 0.5th and 99.5th percentiles, and values outside this range are assigned to the boundary. Simultaneous bands and sup- t tests use 20,000 Gaussian draws. A full fit on the empirical sample (two nuisance functions, five folds, five repetitions) takes about 10 seconds on one CPU core, and re-estimating the final stage with another sieve takes a fraction of a second.

Appendix C. Additional results

Table C.3: Plasmode design: linear TWFE benchmarks.

True θ , estimator	Bias	RMSE	Coverage	Rejection
<i>Moderator: Schooling</i>				
no effect, TWFE-interaction	-0.52	0.75	0.97 (0.01)	0.03 (0.01)
no effect, TWFE-spline	-0.35	1.12	0.98 (0.01)	0.02 (0.01)
linear gradient, TWFE-interaction	-0.47	0.69	0.99 (0.01)	1.00 (0.00)
linear gradient, TWFE-spline	-0.30	1.06	1.00 (0.00)	1.00 (0.00)
threshold, TWFE-interaction	-0.69	0.85	0.98 (0.01)	1.00 (0.00)
threshold, TWFE-spline	-0.22	1.04	1.00 (0.00)	1.00 (0.00)
<i>Moderator: Initial productivity</i>				
no effect, TWFE-interaction	1.28	1.34	0.54 (0.04)	0.47 (0.04)
no effect, TWFE-spline	2.91	3.00	0.10 (0.02)	0.91 (0.02)
linear gradient, TWFE-interaction	1.28	1.35	0.53 (0.05)	1.00 (0.00)
linear gradient, TWFE-spline	2.93	3.03	0.13 (0.03)	1.00 (0.00)
threshold, TWFE-interaction	0.72	0.85	0.98 (0.01)	1.00 (0.00)
threshold, TWFE-spline	2.90	3.01	0.13 (0.03)	1.00 (0.00)
<i>Moderator: Years since take-off</i>				

Notes: see Table 4. TWFE with a linear interaction or with additive spline controls and a spline in the moderator; tercile contrast only. Years since take-off is omitted from the linear benchmarks.

Table C.4: Penalty, identifying variation and treatment support.

Specification	$\hat{\lambda}$	edf	Residual share	Residual share by tercile	Median D by tercile	Within-country s.d. of D	Countries by tercile
A: Baseline	17.8	2.7	0.22	0.11 / 0.31 / 0.29	0.04 / 0.30 / 0.74	0.10 / 0.16 / 0.19	76 / 96 / 72
A: Common sample	1e+04	2.0	0.23	0.10 / 0.36 / 0.32	0.05 / 0.30 / 0.75	0.10 / 0.16 / 0.18	74 / 92 / 70
B: Baseline	1	4.1	0.22	0.14 / 0.35 / 0.23	0.07 / 0.31 / 0.68	0.17 / 0.26 / 0.31	53 / 49 / 47
B: Common sample	56.2	2.2	0.23	0.15 / 0.37 / 0.26	0.07 / 0.33 / 0.69	0.17 / 0.27 / 0.31	51 / 46 / 44
C: Baseline	0.0562	5.4	0.19	0.05 / 0.38 / 0.37	0.02 / 0.30 / 0.80	0.03 / 0.13 / 0.08	142 / 149 / 126

Notes: $\hat{\lambda}$: cross-validated penalty (median over splits; 10^4 is the upper end of the grid, an effectively linear fit); edf: effective degrees of freedom. “Residual share”: variance of the two-way demeaned residual $D - \hat{\mu}$ relative to the variance of two-way demeaned D , overall and by tercile (low/middle/high) of the moderator. Countries by tercile can overlap for time-varying moderators.

Table C.5: Further robustness checks: schooling moderator.

Specification	Average	Tercile of moderator			High –	BLP	Constancy
	(FE-DML)	Low	Middle	High	low	slope	test p
<i>Panel A. Moderator: mean years of schooling</i>							
Original specification	–0.60 (1.14)	–2.08 (2.25)	–0.95 (1.21)	0.29 (1.43)	2.37 (2.85)	0.32 (0.39)	0.48
Adding the investment share	–2.56** (1.30)	–0.92 (2.90)	–2.38* (1.39)	–3.25* (1.67)	–2.34 (3.63)	–0.34 (0.51)	0.97
Adding mobile and broadband, no zero-filling [†]	–1.46 (1.32)	–3.43 (2.19)	–1.36 (1.53)	2.22 (1.48)	5.65* (3.33)	0.63 (0.49)	0.09
Treatment: composite ICT index	–3.07* (1.57)	–1.35 (3.74)	–2.96* (1.77)	–4.27** (1.74)	–2.06 (4.48)	–0.28 (0.61)	0.98
Unpenalised sieve [†]	–2.92** (1.35)	–1.43 (2.73)	–3.19* (1.79)	–2.12 (1.51)	–0.83 (2.99)	–0.36 (0.43)	0.43
Three interior knots [†]	–2.92** (1.35)	–1.99 (2.75)	–2.92** (1.37)	–3.35** (1.63)	–1.07 (3.24)	–0.19 (0.45)	0.79
GCV penalty [†]	–2.92** (1.35)	–1.81 (2.67)	–2.86** (1.45)	–3.03* (1.69)	–1.22 (3.24)	–0.19 (0.46)	0.88
Two-year lag of internet [†]	–2.17* (1.28)	–2.45 (3.29)	–2.16 (1.50)	–1.52 (1.48)	1.17 (4.04)	0.16 (0.55)	0.79
Excluding 2020–2021 [†]	–2.96** (1.29)	–3.51 (3.10)	–3.17** (1.42)	–2.97* (1.65)	0.50 (3.79)	0.07 (0.51)	1.00
Excluding 2025 [†]	–3.46*** (1.30)	–2.11 (2.67)	–3.38** (1.39)	–4.10** (1.71)	–1.83 (3.39)	–0.26 (0.46)	0.76
Untrimmed outcome [†]	–3.33** (1.49)	–2.35 (3.12)	–3.23** (1.53)	–3.89* (2.01)	–1.54 (4.05)	–0.21 (0.55)	0.72
Winsorised outcome [†]	–2.72** (1.32)	–1.77 (2.90)	–2.57* (1.39)	–3.11* (1.69)	–1.12 (3.64)	–0.15 (0.49)	0.78
Without data screening [†]	–2.75** (1.27)	–2.91 (2.68)	–2.79** (1.34)	–2.53 (1.72)	0.38 (3.46)	0.05 (0.47)	0.90
Excluding resource exporters [†]	–1.85 (1.39)	–2.38 (2.64)	–1.82 (1.43)	–1.38 (2.00)	1.32 (3.37)	0.17 (0.45)	0.94
High statistical capacity (SPI) [†]	–2.32* (1.35)	–2.24 (2.58)	–2.76** (1.37)	–1.56 (1.97)	0.44 (3.38)	0.04 (0.64)	0.85
Outcome: GDP per capita growth [†]	–2.75** (1.31)	–2.94 (2.62)	–2.66** (1.32)	–3.54* (1.81)	–0.95 (3.55)	–0.13 (0.48)	0.82
Population-weighted averages [†]	–2.92** (1.35)	–1.99 (2.75)	–2.92** (1.37)	–3.34** (1.63)	–1.06 (3.24)	–0.19 (0.45)	0.79
No Mundlak means [†]	–2.44** (1.09)	–1.28 (1.94)	–2.55** (1.11)	–3.03** (1.42)	–1.24 (2.46)	–0.19 (0.33)	0.85
WDI data, PWT 11.0 countries, 1996–2023 [†]	–3.79*** (1.32)	–8.28*** (3.09)	–4.11*** (1.35)	–1.32 (1.74)	6.97* (3.73)	1.00* (0.54)	0.09
Moderator: learning-adjusted schooling [†]	–2.94** (1.26)	–3.14 (2.88)	–2.99** (1.38)	–2.64* (1.37)	0.73 (3.51)	0.13 (0.60)	0.92
PWT 11.0, 1996–2019 [†]	–1.48 (1.43)	–7.47 (6.55)	–0.81 (2.18)	–0.96 (1.28)	4.74 (6.71)	7.62 (6.06)	0.54
PWT 11.0: outcome TFP growth [†]	–2.80** (1.24)	–10.50*** (4.08)	–3.08* (1.81)	–1.57 (1.47)	8.81* (4.63)	7.63 (5.21)	0.38

Notes: see Table 7. [†]: three repetitions of the sample split. “GCV penalty” selects λ by generalised cross-validation. “Population-weighted averages” changes only the weighting of the sample-average effect, which is reported in the text. “No zero-filling” leaves broadband missing before a country’s first reported value instead of setting it to zero.

Table C.6: Further robustness checks: initial productivity and years since take-off.

	Average	Tercile of moderator			High –	BLP	Constancy
Specification	(FE-DML)	Low	Middle	High	low	slope	test p
<i>Panel B. Moderator: initial log GDP per worker</i>							
Original specification	−0.62 (1.17)	−0.39 (2.38)	−1.08 (1.79)	−0.22 (1.63)	0.91 (3.00)	−0.32 (1.50)	0.46
Adding the investment share	−2.56** (1.30)	−2.06 (2.95)	−3.85** (1.90)	−1.66 (1.47)	0.50 (3.38)	−0.29 (1.45)	0.74
Adding mobile and broadband, no zero-filling [†]	−1.46 (1.32)	−1.38 (2.03)	−2.02 (2.19)	−1.11 (1.96)	0.44 (2.80)	0.18 (1.32)	0.49
Treatment: composite ICT index	−3.07* (1.57)	−5.52* (3.14)	−4.97** (2.37)	−0.23 (1.43)	5.75 (3.52)	2.30* (1.36)	0.08
Unpenalised sieve [†]	−2.92** (1.35)	−3.52 (3.21)	−6.76*** (1.78)	0.82 (1.41)	3.56 (3.59)	1.69 (1.75)	0.03
Three interior knots [†]	−2.92** (1.35)	−0.98 (2.82)	−5.29*** (1.80)	−1.17 (1.38)	−0.28 (3.05)	−0.49 (1.23)	0.16
GCV penalty [†]	−2.92** (1.35)	−1.94 (3.23)	−6.33*** (1.84)	0.62 (1.39)	0.24 (4.18)	−0.31 (2.44)	0.03
Two-year lag of internet [†]	−2.17* (1.28)	−0.40 (2.92)	−2.80* (1.70)	−2.48* (1.37)	−2.08 (3.35)	−0.99 (1.33)	0.62
Excluding 2020–2021 [†]	−2.96** (1.29)	−1.94 (2.52)	−4.21*** (1.63)	−2.28* (1.35)	−0.18 (2.97)	−0.28 (1.16)	0.15
Excluding 2025 [†]	−3.46*** (1.30)	−1.66 (2.90)	−4.58** (1.81)	−3.23*** (1.24)	−1.42 (3.27)	−0.84 (1.31)	0.33
Untrimmed outcome [†]	−3.33** (1.49)	−2.89 (3.32)	−4.81** (1.87)	−2.02* (1.08)	0.87 (3.54)	0.13 (1.42)	0.14
Winsorised outcome [†]	−2.72** (1.32)	−1.06 (2.92)	−4.68*** (1.69)	−2.10* (1.26)	−0.40 (3.24)	−0.57 (1.31)	0.15
Without data screening [†]	−2.75** (1.27)	−1.83 (2.70)	−4.56*** (1.76)	−1.29 (1.37)	0.35 (3.05)	−0.31 (1.22)	0.25
Excluding resource exporters [†]	−1.85 (1.39)	−1.26 (2.40)	−2.27 (2.51)	−0.46 (1.49)	−0.71 (2.99)	−0.42 (1.19)	0.39
High statistical capacity (SPI) [†]	−2.32* (1.35)	−3.12 (2.33)	−2.62 (1.70)	−0.38 (1.22)	2.73 (2.67)	0.52 (1.83)	0.02
Outcome: GDP per capita growth [†]	−2.75** (1.31)	−0.60 (2.75)	−4.21** (1.71)	−2.18 (1.90)	−0.48 (3.30)	−0.25 (1.46)	0.65
Population-weighted averages [†]	−2.92** (1.35)	−1.00 (2.80)	−5.36*** (1.80)	−1.15 (1.38)	−0.25 (3.06)	−0.47 (1.24)	0.15
No Mundlak means [†]	−2.44** (1.09)	−0.13 (2.21)	−5.51*** (1.69)	0.46 (1.22)	0.77 (2.48)	−1.10 (1.26)	<0.01
Moderator: distance to US frontier [†]	−2.00 (1.25)	3.32 (2.79)	−4.26** (1.99)	−0.96 (1.14)	−4.73 (2.97)	−2.82* (1.48)	0.16
Countries with pre-sample productivity only [†]	−2.76** (1.28)	−1.84 (2.71)	−4.12** (1.66)	−1.74 (1.49)	0.65 (2.95)	−0.25 (1.27)	0.36
PWT 11.0, 1996–2019 [†]	−1.48 (1.43)	−6.92 (4.71)	−2.69 (1.85)	0.62 (1.63)	7.24 (5.12)	3.23 (2.29)	0.47
PWT 11.0: outcome TFP growth [†]	−2.80** (1.24)	−6.41** (3.07)	−4.00*** (1.34)	−1.37 (1.72)	4.13 (3.25)	1.85 (1.50)	0.67
PWT 11.0, national-prices moderator [†]	−3.34*** (1.29)	−10.05*** (3.12)	−2.75** (1.37)	−0.78 (1.35)	9.28** (3.78)	4.22** (1.74)	0.06
PWT 10.0, national-prices moderator [†]	−1.74 (1.60)	−10.70*** (3.66)	−2.85** (1.36)	0.81 (1.62)	11.50*** (4.24)	5.39*** (1.86)	0.03
<i>Panel C. Moderator: years since internet take-off (share ≥ 10%)</i>							
Pre-COVID sample, 1996–2019 [†]	−1.94 (1.70)	−9.49** (4.48)	−4.28* (2.40)	−0.58 (2.06)	8.91* (5.14)	0.44* (0.25)	0.24
Adding mobile and broadband [†]	−1.42 (1.44)	16.28 (11.23)	−2.17 (1.83)	−0.77 (1.59)	−16.71 (11.34)	−1.08* (0.58)	0.05

Notes: see Table 7. [†]: three repetitions of the sample split. “GCV penalty” selects λ by generalised cross-validation. “Population-weighted averages” changes only the weighting of the sample-average effect, which is reported in the text. “No zero-filling” leaves broadband missing before a country’s first reported value instead of setting it to zero.

References

- Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32.
- Cameron, A.C., Miller, D.L., 2015. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources* 50, 317–372.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J., 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* 21, C1–C68.
- Gaure, S., 2013. OLS with multiple high dimensional category variables. *Computational Statistics & Data Analysis* 66, 8–18. doi:10.1016/j.csda.2013.03.024.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree, in: *Advances in Neural Information Processing Systems*, pp. 3146–3154.